LOCALISING TWICE IS THE LOCALISATION AT THE IMAGE OF THE FIRST MULTIPLICATIVE SUBSET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The universal property of the localisation. Used twice in the counted proof (lines 1499-1504 and 1507-1510). Produces both maps out of the iterated localisation without computing on fractions first.
- The image of a multiplicative subset as a multiplicative subset. Named in the statement and used in the construction of the first map. Supplies the subset whose elements must be inverted to invoke the universal property.

Proposition 0.1. *Let $\overline{S}$ be the image of S in $S'^{-1}A$, then $(SS')^{-1}A$ is isomorphic to $\overline{S}^{-1}(S'^{-1}A)$.*

Proof. The map sending $x \in A$ to $x/1 \in (SS')^{-1}A$ induces a map sending $x/s \in S'^{-1}A$ to $x/s \in (SS')^{-1}A$, by universal property. The image of the elements in $\overline{S}$ are invertible in $(SS')^{-1}A$. By the universal property we get a map $f : \overline{S}^{-1}(S'^{-1}A) \to (SS')^{-1}A$ which maps $(x/s')/(s/1)$ to x/ss'.

On the other hand, the map from A to $\overline{S}^{-1}(S'^{-1}A)$ sending $x \in A$ to $(x/1)/(1/1)$ also induces a map $g : (SS')^{-1}A \to \overline{S}^{-1}(S'^{-1}A)$ which sends x/ss' to $(x/s')/(s/1)$, by the universal property again. It is immediately checked that f and g are inverse to each other, hence they are both isomorphisms. □

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